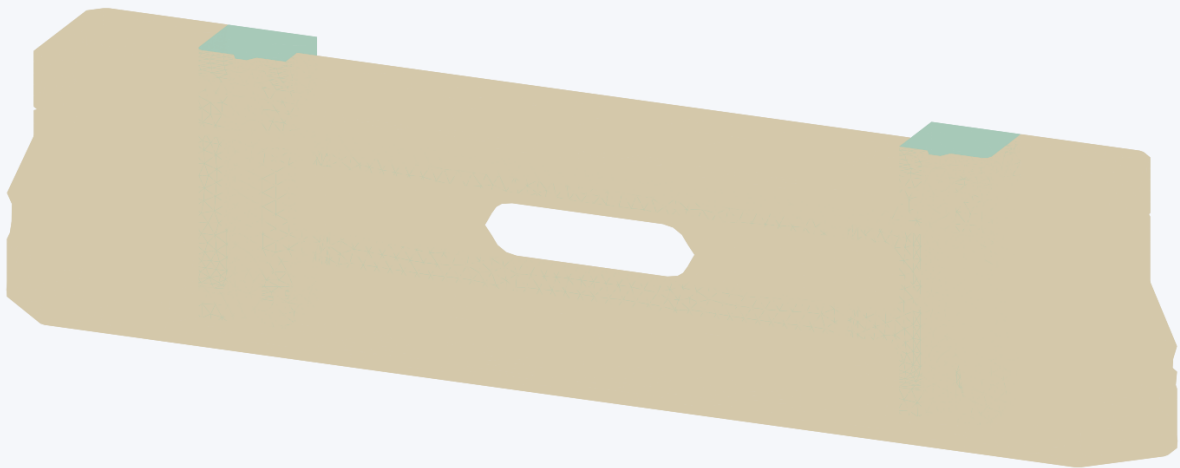


Selected-run evidence status: stopped\_early

## Geometry

### Undeformed geometry

Exterior shape from the selected mesh



Mesh-derived exterior • source geometry unchanged

Undeformed geometry from selected mesh. Exterior shape derived from the selected run's mesh; this is not a CAD render.

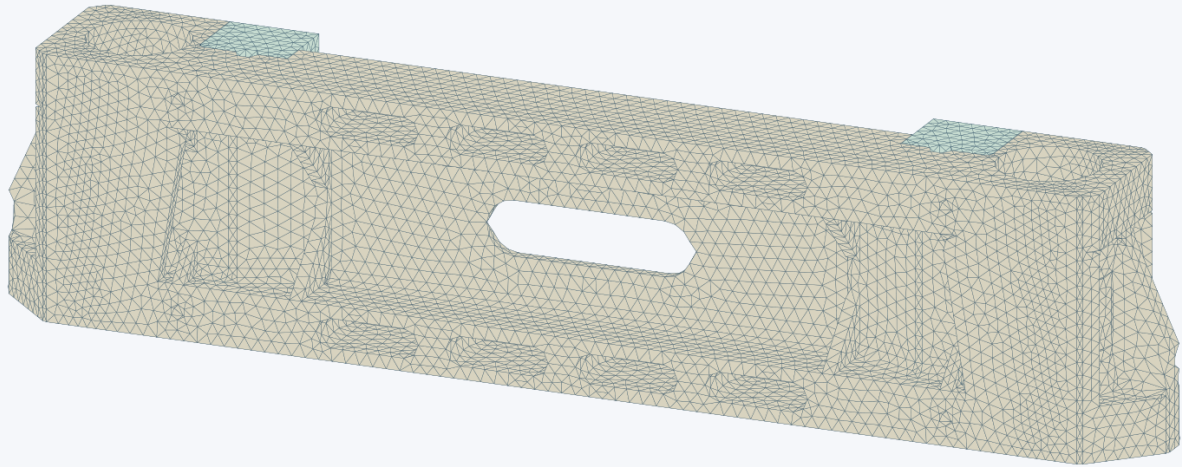
## Mesh

### Mesh overview and local detail

|                            |        |     |
|----------------------------|--------|-----|
| Mesh nodes                 | 66280  |     |
| Tetrahedral elements       | 311537 |     |
| Meshed parts               | 2      |     |
| Nominal element size       | 4.0    | mm  |
| Minimum mean-ratio quality | 0.0268 | 0-1 |

## Mesh overview

Undeformed exterior mesh

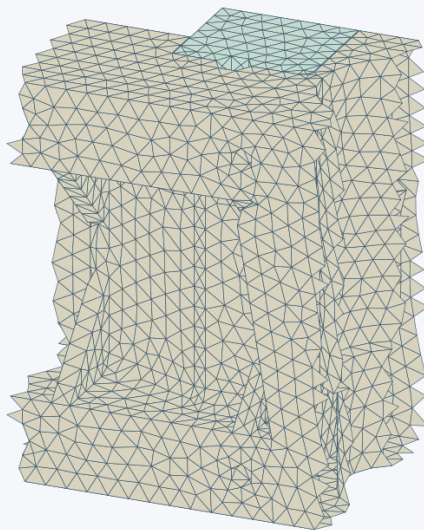


Exterior mesh faces • undeformed geometry

Mesh overview. Undeformed exterior topology of the selected case, verified against its recorded solved-mesh hashes.

## Local mesh detail

Local mesh detail



Exterior mesh faces • undeformed geometry

Local mesh detail. Local view of the selected mesh; no local refinement claim is made.

Mean-ratio quality describes element shape and does not establish stress or time-step convergence.

## Setup

Recorded selected-case model

|                                 |           |               |
|---------------------------------|-----------|---------------|
| Initial impact velocity         | 50        | m/s           |
| Requested simulation duration   | 8         | ms            |
| Foam depth                      | 203.2     | mm            |
| Foam width                      | 41.275    | mm            |
| Foam height                     | 152.4     | mm            |
| Foam element size               | 10        | mm            |
| Initial contact gap             | 0.1       | mm            |
| Contact friction coefficient    | 0.1       | dimensionless |
| Foam curve scale                | 1         | dimensionless |
| Time-step scale                 | 0.6       | dimensionless |
| Requested state-output interval | 0.01      | ms            |
| Startup mass-scaling DT2MS      | -1.5e-07  | s             |
| Termination time-step ratio     | 0.01      | dimensionless |
| Motor mass                      | 0.1616    | kg            |
| Moving mass                     | 0.6503408 | kg            |
| Pusher mass                     | 0.4887408 | kg            |
| Initial kinetic energy          | 812.926   | J             |

## Recorded material, contact and boundary assumptions

- Body and bolted cable cap flexible; four local perfectly clamped bolt collars. No bolt preload, slip or failure.
- Motor loaded upper-bound inertia distributed over actual bore nodes; no motor housing stiffness or exact inertia tensor.
- PA12-CF10 elastic isotropic proxy  $E=1806.6\text{MPa}$   $\nu=.35$  density1060; print +Z motor axes. No strength/failure claims.
- No fracture, erosion, motor thrust, carrier, released endcap or invented 35g hardware allowance.
- Rigid 8mm aluminum spreaders initially stationary; fixed x,y and rotations, free z without travel limit.
- Front and rear pusher bore-quadrant averages follow fixed zero-clearance coaxial guide axes through eight disjoint LINEAR\_GLOBAL X/Y equations; every Z DOF is free. Cable compliance and clearance are idealized.

- All axial stopping reaction passes through transient contact; no axial guide constraint.
- Foam rear z fixed, x/y free: frictionless fixed backing; sides unconfined.
- MAT057 synthetic loading and unloading; HU=.05 BETA=0 DAMP=.1 SHAPE=1 FAIL=0. Not a product material model.
- Generic foam anchored at one catalog point; remaining curve chosen for nonbottoming comparative stop.
- Startup-only local mass scaling requested; added mass and inertia error MUST be assessed from solver output.
- Mesh, timestep, energy and momentum require independent assessment.

## Solve

|                                    |                              |    |
|------------------------------------|------------------------------|----|
| Recorded run status                | stopped_early                |    |
| Recorded reached simulation time   | 0.237953                     | ms |
| Recorded requested simulation time | 8                            | ms |
| Recorded solver wall time          | 122.047                      | s  |
| Recorded termination state         | normal_termination_confirmed |    |

## Separate solve charts

## Contact force resultant

No extracted history for this result in the selected run.

Contact force resultant. No extracted source available.

## Mass-weighted forward velocity

No extracted history for this result in the selected run.

Mass-weighted forward velocity. No extracted source available.

## Moving assembly travel

No extracted history for this result in the selected run.

Moving assembly travel. No extracted source available.

## Foam compression

No extracted history for this result in the selected run.

Foam compression. No extracted source available.

## Regional P99 von Mises stress

No extracted history for this result in the selected run.

Regional P99 von Mises stress. No extracted source available.

## Global energy

No extracted history for this result in the selected run.

Global energy. No extracted source available.

## Results

# Run 008 - impact evidence

**Evidence status:** stopped\_early. This report describes only extracted evidence from the selected run. Numerical checks alone do not establish physical qualification.

## Geometry and model setup

| Recorded model item | Value     |
|---------------------|-----------|
| kind                | impact    |
| pusher kg           | 0.488741  |
| body kg             | 0.41625   |
| cable cap kg        | 0.0724904 |
| motors kg           | 0.1616    |
| moving kg           | 0.650341  |
| initial ke J        | 812.926   |
| spreader each kg    | 0.135871  |

## Solved mesh

Recorded mesh source: mesh\_cap\_nested\_longest. Mesh and timestep convergence are not established by these histories.

## Motion and stopping event

Signed forward velocity, travel and foam compression are measured histories. Negative forward velocity denotes rebound; a partial history cannot establish the complete stopping event.

| Measured result | Value | Unit |
|-----------------|-------|------|
|-----------------|-------|------|

## Regional stress and energy

Regional P99 values are elastic demand measures across saved states, not printed-part strength predictions. Force peaks and percentile peaks can occur at different times.

## Numerical checks and qualification

No selected-run numerical check summary is available.

No extracted results are available for this run yet.

## Images, animations and source evidence

- execution.json
- model.json
- startup\_mass\_audit.json

Generated from selected-run extracted evidence; no authored report is available. Animation figures and their captions are excluded from this PDF; animations remain available in the standalone HTML.